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Ultrasound-assisted metabolite detection in different extraction processes of *Bletilla striata* and bitter metabolite detection

Juan Zhou [a,b,1], Yushen Feng [a,1], Wenhao Zhou [a,1], Mengying Zhang [a], Fugui Liu [a], Jian Mao [c], Dajun Wu [d], Yunpeng Cao [e,*], Yigao Wu [a,f,*], Lan Jiang [a,b,*]

[a] *Anhui Province Key Laboratory of Non-coding RNA Basic and Clinical Transformation (Wannan Medical College), Central Laboratory, Yijishan Hospital of Wannan Medical College, Wuhu 241000, China*
[b] *Central Laboratory, Fuzhou University Affiliated Provincial Hospital, Fuzhou 35000, China*
[c] *Yangtze River Delta Information Intelligence Innovation Research Institute, Wuhu 241000, Anhui, China*
[d] *Anhui Runhua Ecological Forestry Co., Ltd., Guangde 242200, Anhui, China*
[e] *CAS Key Laboratory of Plant Germplasm Enhancement and Specialty Agriculture, Wuhan Botanical Garden, Chinese Academy of Sciences, Wuhan 430074, Hubei, China*
[f] *Department of Medical Psychology, Yijishan Hospital of Wannan Medical College, Wuhu 241000, Anhui, China*

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ABSTRACT

Bletilla striata, a medicinal orchidaceous plant, is recognized for its significant pharmacological value. However, the lack of comparative metabolomic data across different extraction methods for analyzing its bioactive components has significantly undervalued the application potential of *B. striata* in the traditional Chinese medicine market. Using six ultrasound-assisted extraction methods and UPLC-MS/MS, this study identified 1,945 metabolites in *B. striata* extracts. The dominant categories were lipids (51.35%), flavonoids (18.00%), and phenolic acids (12.51%). KEGG analysis revealed alterations in flavonoids and isoflavonoids biosynthesis pathways. Thirteen bitter metabolites, including cinnamic acid, were identified in *B. striata* tubers, underscoring their potential pharmacological applications, such as anti-inflammatory, antioxidant and antibacterial activities. Optimizing different extraction methods can better preserve the bioactive components of *B. striata* extracts, thereby enhancing its potential applications in the food and pharmaceutical industries.

1. Introduction

Plant-derived extracts encompass a range of bioactive compounds, such as phytochemicals, crude extracts, and essential oils, meticulously obtained from diverse plant sources[1]. The in vitro investigation into the antimicrobial and synergistic properties of these extracts has garnered increased attention. The extracts and phytochemicals from *Cassia alata* against some clinical isolates of MDR bacteria[2]. Extracts from *Roselle flowers* have been traditionally utilized in medicine to address conditions such as fever, hypertension, and hepatic diseases [3]. The treatment with *Angelica keiskei* extract significantly reduced blood glucose levels in rats fed a high-fructose diet [4]. The combined extracts of lemon and pepper residue[5], along with the by-products of olive, grape, and lemon[6], exhibits strong antioxidant activity. The aqueous extracts of cinnamon is rich in bioactive compounds, providing substantial protection against cancer and inflammatory conditions[7]. Additionally, the aqueous extracts of *Lavandula stoechas* [8] and extracts from *Angelica keiskei* [9] show anti-diabetic properties in mouse models.

Metabolomics, which seeks to comprehensively analyze all metabolites in biological samples, holds significant potential for elucidating plant metabolic processes [10]. In recent years, this approach has emerged as a potent tool for investigating alterations in metabolite diversity and content resulting from various processing treatments [11]. Significant disparities were observed in the metabolite composition between fresh and dried Chinese herbs [12]. The process of drying citronella has been observed to augment the flavonoid content while

* Corresponding authors at: Anhui Province Key Laboratory of Non-coding RNA Basic and Clinical Transformation (Wannan Medical College), Central Laboratory, Yijishan Hospital of Wannan Medical College, Wuhu 241000, China (L. Jiang, Y. Wu); CAS Key Laboratory of Plant Germplasm Enhancement and Specialty Agriculture, Wuhan Botanical Garden, Chinese Academy of Sciences, Wuhan 430074, Hubei, China (Y. Cao).

E-mail addresses: xfcypeng@126.com (Y. Cao), wuyigao2000@126.com (Y. Wu), jianglanhi@163.com (L. Jiang).

[1] These authors contributed equally to this work.

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simultaneously diminishing the levels of amino acids, organic acids, and vitamins [13]. The method of extraction, whether through cold or hot processes, significantly impacts the metabolites present in plant extracts [14]. The cold water extract of the *Italian wax* flower is rich in caffeic acid derivatives, including chlorogenic acid and aromatic hydroxybenzoic acid. In contrast, the hot water extract is more effective at extracting total phenols and specific flavonoids, thereby enhancing the antioxidant properties of the extract [15]. These differences not only influence the pharmacological effects of the herbal medicines but may also significantly impact their clinical applications and the manifestation of their medicinal effects[16]. Therefore, a comprehensive discussion of these differences is crucial for optimizing the extraction process of herbal medicines and enhancing their medicinal efficacy[17].

B. striata is a traditional Chinese medicine (TCM) that is both medicinal and edible, known for its richness in *B. striata* polysaccharides (BSP) [18]. Typically, dried tuber slices are used in TCM formulations, while by-products and scraps are utilized to extract BSP [19]. In Yunnan, China, fresh *B. striata* is also used to prepare stew, which has a slightly bitter taste [20]. The production *of B. striata* tubers is described in the Chinese Pharmacopoeia as follows: fresh tubers are boiled or steamed and then dried through solar exposure [21]. According to the 2015 edition of the Chinese Pharmacopoeia, *B. striata* possesses several medicinal properties, including hemostasis, anti-inflammation, and the promotion of tissue regeneration and organ function. It is used in the treatment of wounds, as well as conditions such as hematemesis, hemoptysis, and edema [22]. Research on the pharmacological properties of *B. striata* extracts has demonstrated various bioactivities, including antibacterial and antitumor effects, hemostatic properties, wound healing promotion, immunomodulation, and the enhancement of bone marrow hematopoiesis [23]. Using the UPLC-MS/MS non-targeted metabolomics approach, metabolites were extracted from both fresh *B. striata* tubers and dried *B. striata* tubers through cold and hot extraction methods. This analysis revealed distinct metabolite profiles, providing valuable insights into the medicinal and nutritional potential of *B. striata* tubers. These findings offer novel opportunities for drug development and the treatment of various diseases. Bitter compounds primarily serve as a defense mechanism against external threats and possess extensive medicinal value. For instance, cucurbitacin, known for its bitter taste, exhibits antioxidant, anti-inflammatory, antibacterial, and potential antitumor properties [24]. Regular consumption of bitter beverages, such as coffee and tea, has been associated with a reduced risk of type 2 diabetes and cardiovascular diseases [25,26].

2. Materials and methods

2.1. Plant materials

Three-year old fresh *B. striata* tubers were procured from Wenshan, Yunnan Province, while dried *B. striata* powder were sourced from Bozhou, Anhui Province, China (Originally cultivated in Wenshan, Yunnan). Both the fresh and dried *B. striata* tubers were juiced to obtain *B. striata* extracts (BSE) using cold and hot extraction methods, respectively. The extracted samples were preserved for metabolomics analyses, with three biological replicates included for each of the Fresh *B. striata* tubers (FBST), Dried *B. striata* tubers (DBST), Fresh *B. striata* cold (FBSC), Dried *B. striata* cold (DBSC), Fresh *B. striata* hot (FBSH), and Dried *B. striata* hot (DBSH) samples.

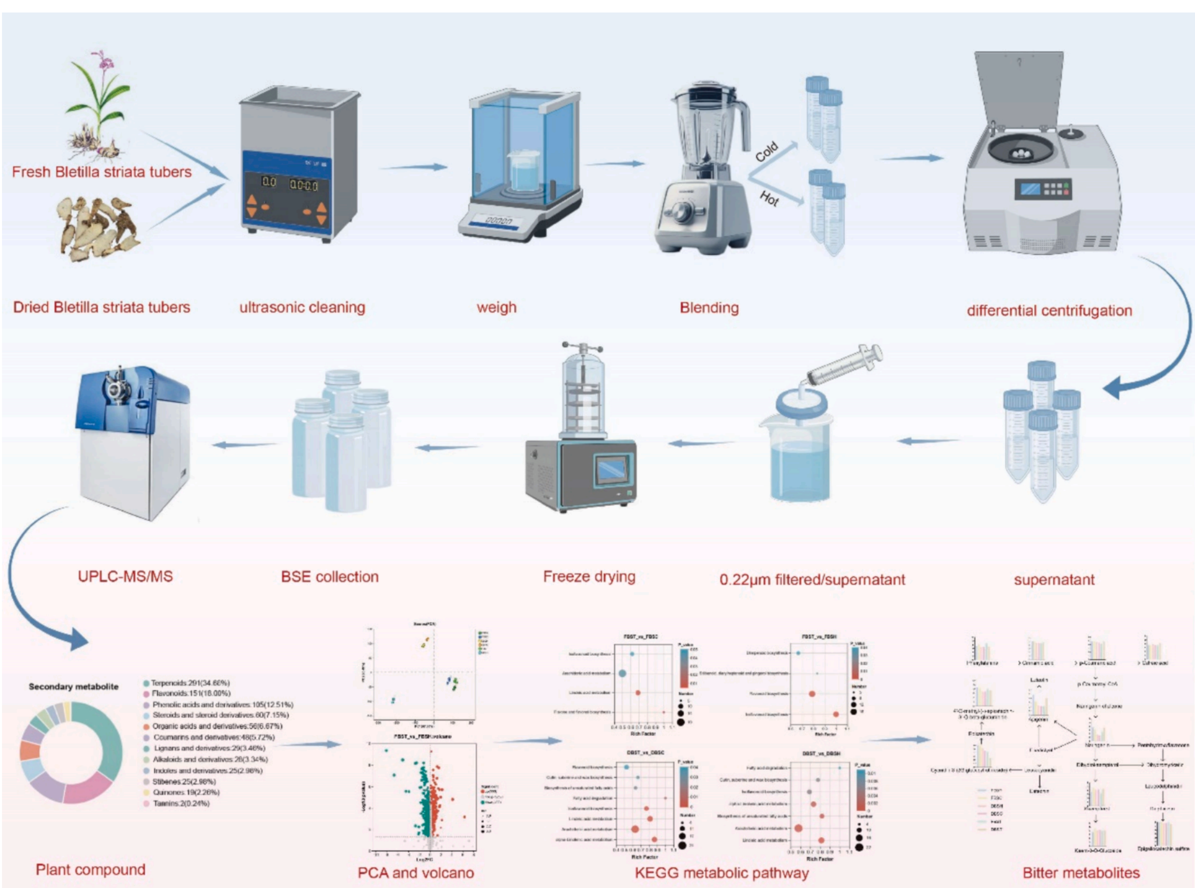

Fig. 1. Specific workflow diagram.

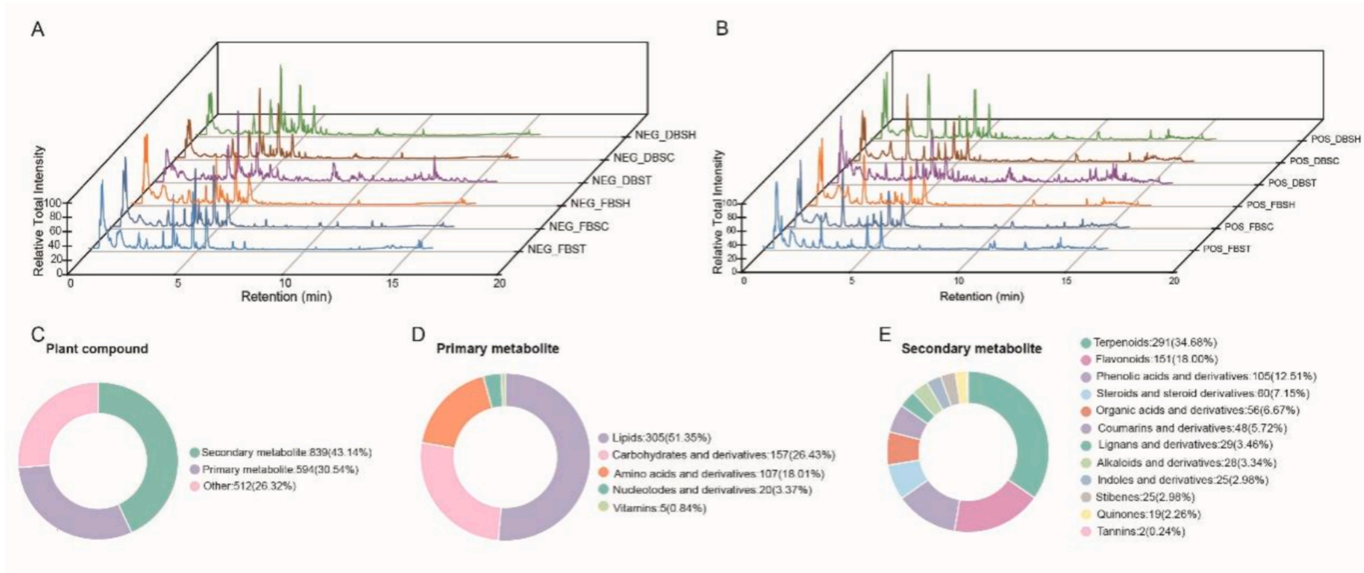

Fig. 2. Metabolic characteristics of BSE. The base peak chromatograms of BSE are presented in negative (A) and positive (B) ion modes. Classification of plant compound (C), primary metabolite (D) and secondary metabolite (E). NEG, negative; POS, positive; FBST, fresh *B. striata* tubers; DBST, dried *B. striata* tubers; FBSC, fresh *B. striata* cold; DBSC, dried *B. striata* cold; FBSH, fresh *B. striata* hot; DBSH, dried *B. striata* hot.

2.2. Ultrasound-assisted extraction

A precise 10 g sample of dried *B. striata* tubers was weighed into a beaker, followed by the addition of 400 mL ddH$_2$O. In a separate beaker, 100 g of fresh *B. striata* tubers were combined with 400 mL ddH$_2$O and placed in an ultrasonic bath (SN-QX-20D, Shanghai Shangpu Instrument Co., Ltd., Shanghai, China), which served as the ultrasonic source for the experiment. Ultrasonic extraction was performed for 20 min, with an ultrasonic power of 350 W and an operating frequency of 40 kHz. The process alternated between 10 s of sonication and 10 s of rest in a cyclic pattern [27]. The temperature was maintained at a constant 5 °C throughout the extraction. Following the ultrasound-assisted extraction, juice extraction was carried out.

2.3. Preparation of BSE

Fresh *B. striata* after cleaning using ultrasound-assisted. A quantity of 100 g was then taken and cut into small pieces for subsequence. Under cold (−4 °C −0 °C) and hot (80 °C −100 °C) temperatures, juice with an ultrasonic wall breaker, filter the gauze, centrifuge at 1000 g for 10 min, 3000 g for 30 min, 10,000 g for 30 min, take the supernatant, and finally filter it with a 0.22 μM filtration membrane to obtain BSE, freeze-dry it into powder, and store it at room temperature. Dried *B. striata* tubers take the same approach.

2.4. UPLC-MS/MS measure of BSE

In the study of BSE, metabolomics analysis was conducted on three biological replicates from each group. The methodology for metabolite extraction and UPLC-MS/MS followed established protocols [28,29]. In summary, a precise 100 μL of liquid sample was extracted using ultrasonic methods with an additional 100 μL of liquid sample in a 1.5 mL centrifuge tube. This was followed by the addition of 800 μL of a solution (acetonitrile: methanol = 1:1 (v: v)) containing the internal standard (0.02 mg/mL-2-chloro phenylalanine, among others) for 30 min at 5℃ and 40KHz. The mixture was then incubated at −20℃ for another 30 min. Following this, the mixture was centrifuged at 13,000 × g for 15 min at 4℃. Finally, 200 μL of the supernatant was transferred to a new vial for UPLC-MS analysis. Metabolite identification was performed using Thermo Vanquish LC-MS (Thermo Fisher Scientific, Waltham, MA, USA) according to a previous report[30].

2.5. Metabolomic analysis

The raw data were processed using Progenesis QI 2.3 (Waters Corporation, Milford, CT, USA) for baseline filtration, peak identification, integration, retention time correction, peak alignment, and normalization[31]. Metabolomic data was analyzed using Majorbio platform (cloud.majorbio.com)[32]. The R package "ropls" (Version 1.6.2) was employed to perform principal component analysis (PCA) and orthogonal least partial squares discriminant analysis (OPLS-DA), in conjunction with a 7-cycle interactive validation to evaluate model stability. The Venn and volcano plots are referenced from previously published articles [18,33–35]. Enrichment analysis was conducted using the KEGG database (https://www.genome.jp/kegg/) to identify specific metabolites within functional nodes [18]. The graphic abstract was generated by BioGDP.com[36].

3. Results and discussion

3.1. Overview of the metabolites identified in B. striata extract

To better illustrate the differences in metabolites between the BSE obtained through different extraction processes, the overall metabolic characterization of fresh and dried *B. striata* was performed using UPLC-MS/MS. This analysis utilized an information-dependent acquisition method combined with dynamic background subtraction, mass defect filtration, and neutral loss data acquisition functions. The complete flow chart is presented as a graphical abstract (Fig. 1).

Representative total ion chromatograms of BSE, based on mass-to-charge ratio (M/Z), retention time (RT), and intensity of negative (Fig. 2**A**) and positive (Fig. 2**B**) ions modes, were generated using the UPLC-MS/MS technique. These chromatograms confirmed that the overall status of our detection instrument was within reliable limits. To intuitively present the classification of all compounds in the BSE, pie charts were created to depict the plant compounds, primary metabolites, and secondary metabolites categories (Fig. 2**C-E**). A total of 1945 metabolites, categorized into 18 classes, were identified in the BSE. The primary metabolites comprised lipids: 305(51.35 %), carbohydrates and their derivatives:157(26.43 %), amino acids and their derivatives:107(18.01 %),

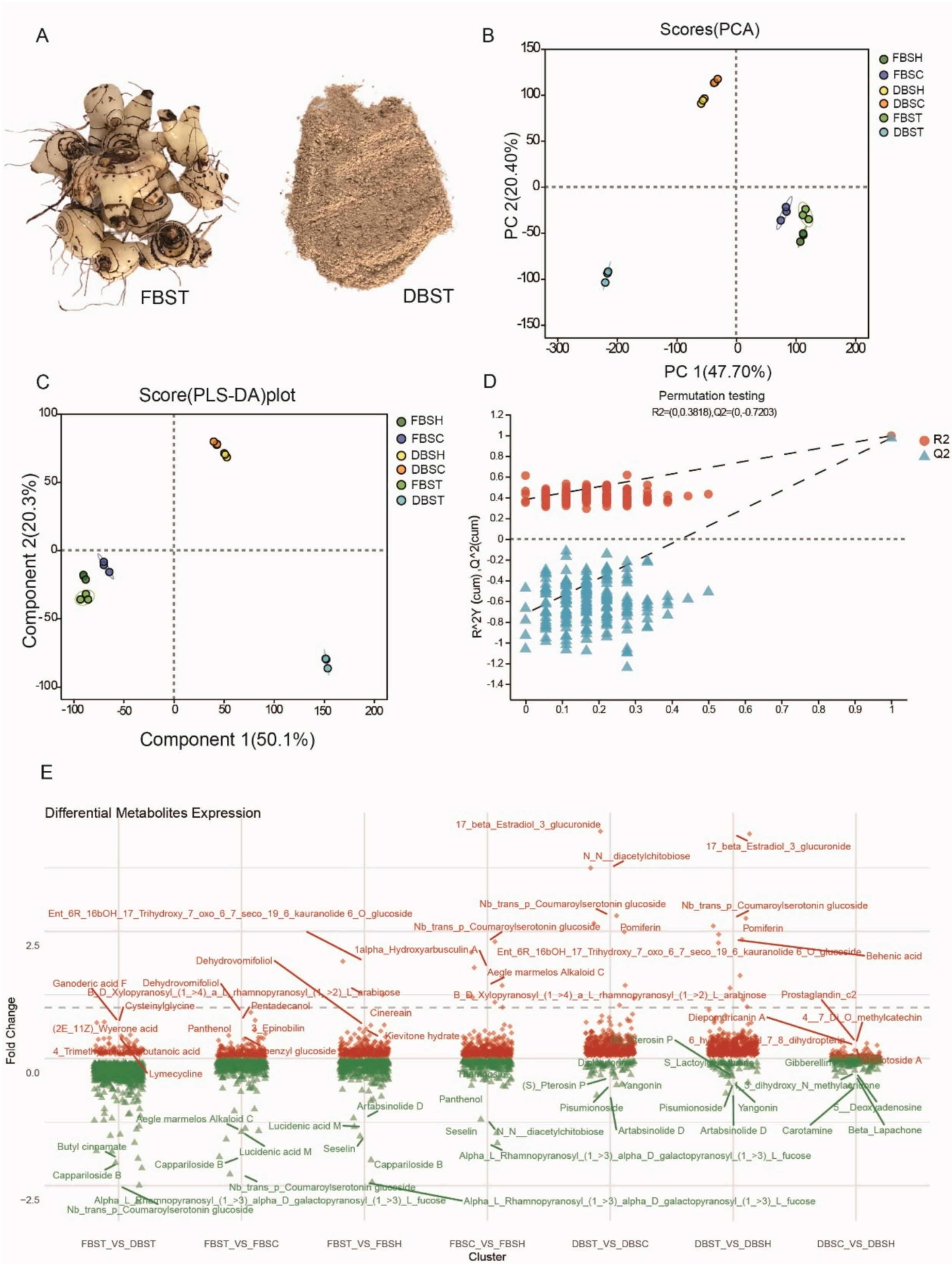

Fig. 3. Multivariate statistical analysis of B. striata extracts. (A)Fresh *B. striata* tubers (left), dried *B. striata* tubers (powder, right). (B) PCA score plots and (C) OPLS-DA score plots of BSE from six different processes. (D) 200 permutation experiments of OPLS-DA models for the BSE from six different processes. (E) Volcano plots of BSE compared between groups, respectively. FBST, fresh *B. striat*a tubers; DBST, dried *B. striata* tubers; FBSC, fresh *B. striata* cold; DBSC, dried *B. striata* cold; FBSH, fresh *B. striata* hot; DBSH, dried *B. striat*a hot.

nucleotides and their derivatives: 20(3.37 %), and vitamins: 5(0.84 %) (Table S1). Lipids play essential physiological roles, including membrane synthesis, energy storage, and the formation of signaling molecules [37]. Medicinal plant extracts are rich in lipid metabolites and have a good protective effect in the treatment and management of skin diseases[38]. BSE is also rich in lipid metabolites (51.35 %), which may play an important role in skin wound healing and antioxidant[39]. Many of these amino acids and their derivatives play a pivotal role in plant development and in mediating responses to environmental stresses[40]. Amino acids are the precursors for numerous primary and secondary metabolic products and play a vital role in promoting human health [41], 107 amino acids and their derivatives were identified in BSE.

Secondary metabolites with bioactivity play an important role in medicinal plants[42]. BSE also contains various bioactive secondary metabolites (**Table S2**), which may contribute to its application in TCM and modern herbal medicine. These secondary metabolites are the material basis of its clinical efficacy and are important indicators for evaluating its quality. Secondary metabolites include terpenoids:291 (34.68 %), flavonoids:151(18.00 %), phenolic acids and derivatives:105 (12.51 %), steroids and steroid derivatives:60(7.15 %), organic acids and derivatives:56(6.67 %), coumarins and derivatives:48(5.72 %), lignans and derivatives:29(3.46 %), alkaloids and derivatives:28(3.34 %), stilbenes: 25(2.98 %), indoles and derivatives:25(2.98 %), quinones:19(2.26 %), and tannins: 2(0.24 %). Flavonoids were the second abundant compounds in BSE. Many flavonoids, for example, quercetin, have biological activities that can contribute to human health[43]. Flavonoids make up about two-thirds of dietary phenolics[44]. The third most abundant metabolite identified in BSE was Phenolic acids and their derivatives. There is increasing awareness and interest in the antioxidant activity and possible health benefits of phenolic acids, which account for almost all of the remaining third[45]. Total of 512 other compounds were identified in BSE (**Table S3**). We selectively grouped the types and amounts of the main six metabolites (**Fig. S1**), phenolic acids and derivatives (**Fig. S1A**), amino acids and derivatives (**Fig. S1B**), terpenoids (**Fig. S1C**), flavonoids (**Fig. S1D**), carbohydrates and derivatives (**Fig. S1E**), lipids (**Fig. S1F**) to plot bar charts (**Table S4**).

3.2. Metabolic footprints of the BSE from different extraction processes

PCA and OPLS-DA were employed to visualize the chemical dynamic changes of metabolites and screen for differential markers[46]. Metabolites were considered significantly different if they exhibited a Variable Importance in Projection (VIP) greater than 1 and a P-value < 0.05, as determined by the VIP from the OPLS-DA model and the P-value derived from Student's *t*-test [47]. PCA highlighted differences between FBST from Yunnan (left) and DBST from Anhui (right) (Fig. 3**A**), revealing variations in BSE composition due to extraction methods. By performing PCA analysis on the samples, we can determine the overall metabolite differences and the degree of variability between samples in different groups [47]. The grouping ellipses of metabolites in BSE from six

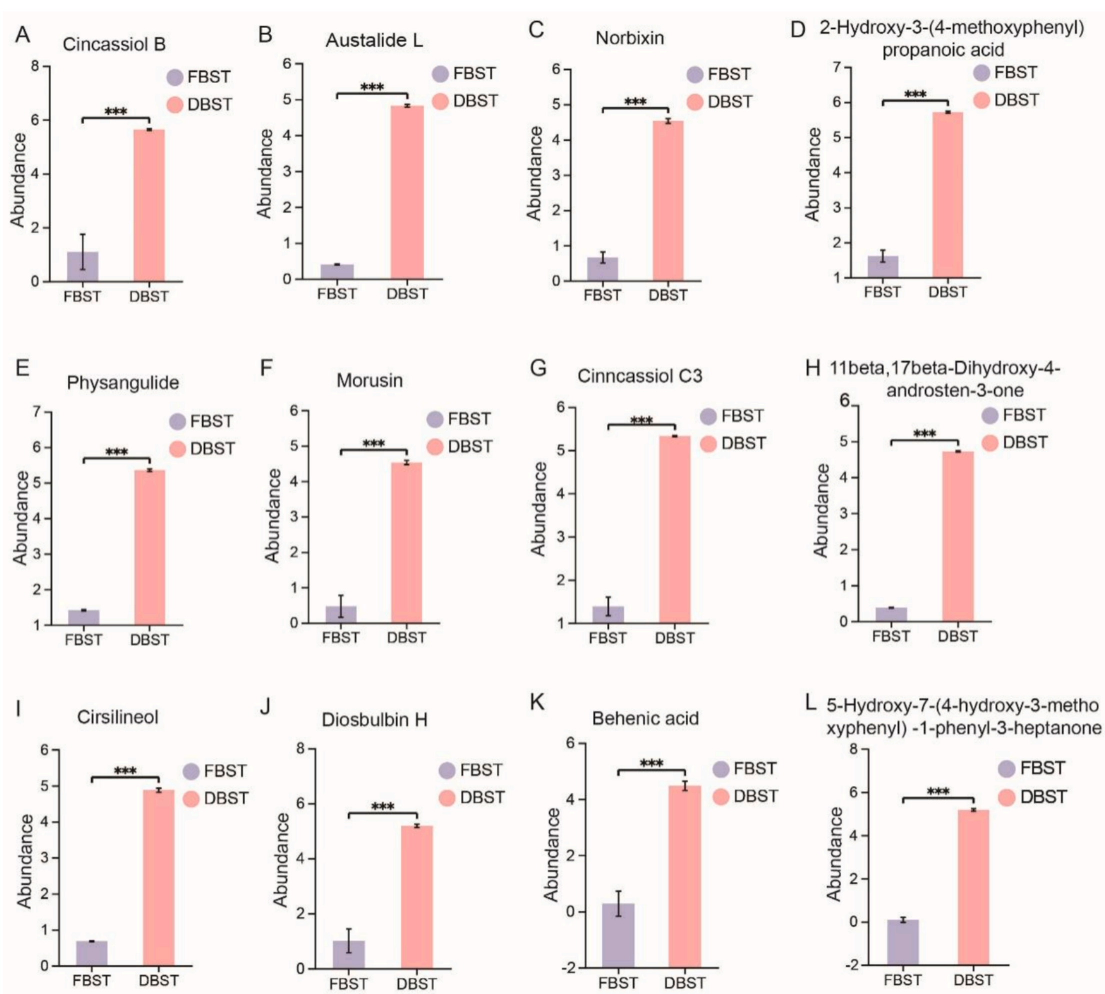

Fig. 4. The bar graphs illustrate that the significantly different compounds between the FBST and DBST groups. (A) Cincassiol B; (B)Austalide L; (C)Norbixin; (D)2-Hydroxy-3-(4-methoxyphenyl) propanoic acid; (E)Physangulide; (F)Morusin; (G)Cinncassiol C3; (H)11beta,17beta-Dihydroxy-4-androsten-3-one; (I)Cirsilineol; (J) Diosbulbin H; (K)Behenic acid; (L)5-Hydroxy-7-(4-hydroxy-3-methoxyphenyl)-1-phenyl-3-heptanone.

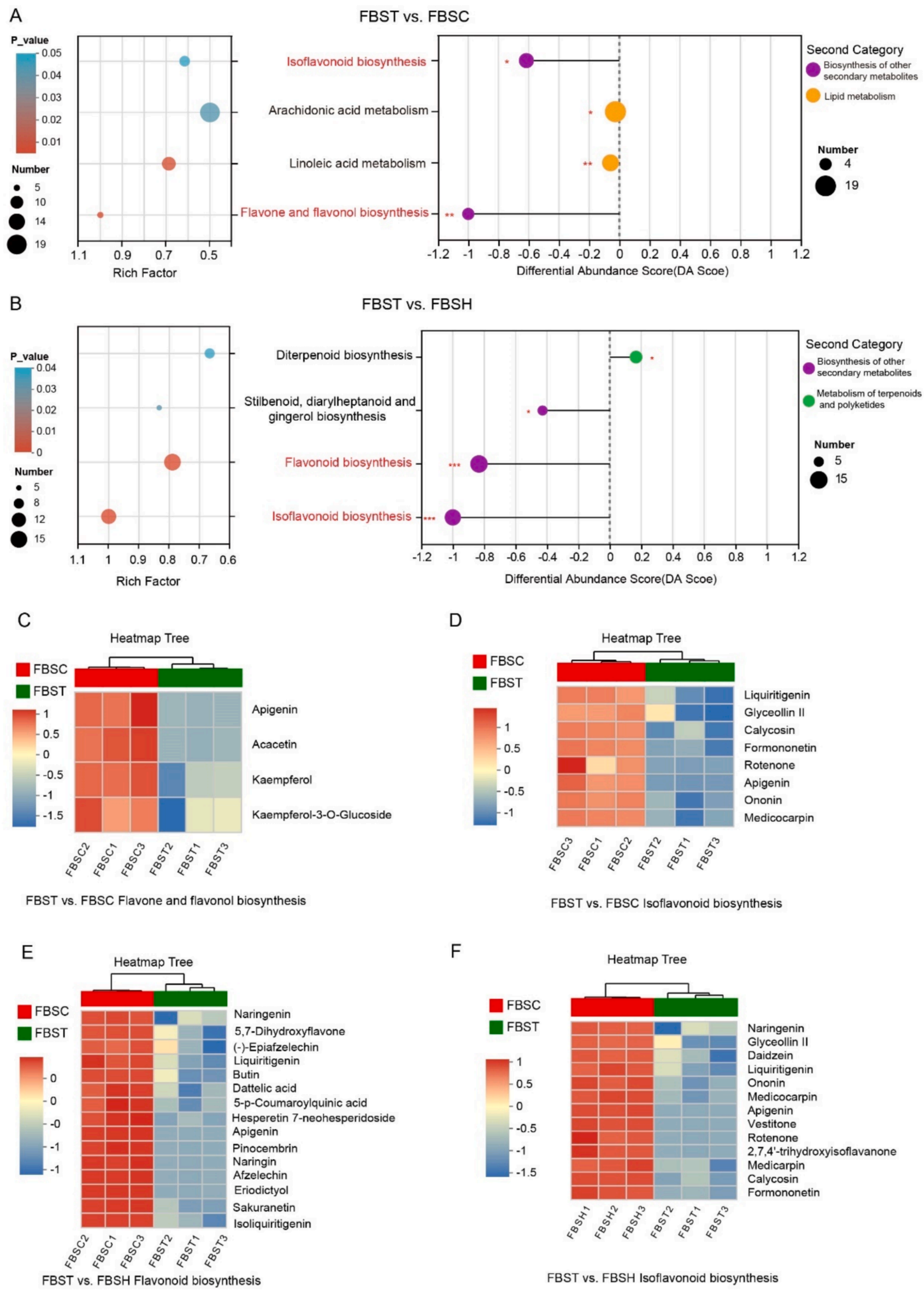

Fig. 5. KEGG enrichment analysis was performed, followed by the visualization of metabolite expression using heatmap. KEGG enrichment analysis (left) and DA score plot (right) of (A) FBST vs. FBSC and (B) FBST vs. FBSH. Heatmap of metabolites in the (C) flavone and flavonol biosynthesis pathway, (D) isoflavonoid biosynthesis pathway for FBST vs. FBSC; and (E) flavone and flavonol biosynthesis pathway, (F) isoflavonoid biosynthesis pathway for FBST vs. FBSH. FBST, fresh *B. striata* tubers; DBST, dried *B. striata* tubers; FBSC, fresh *B. striata* cold; purple means Biosynthesis of other secondary metabolites, green means Metabolism of terpenoids and polyketides, yellow means lipid metabolism. *, P-value < 0.05, **, P-value < 0.01, ***, P-value < 0.001.

processes were separated (Fig. 3**B-3C**). PCA and OPLS-DA revealed significant differences between DBST compared to cold and hot extracts, although FBST clustered in the same quadrant as the extracts (Fig. 3**B-3C**). OPLS-DA was performed to identify metabolites responsible for differences among the six groups (Fig. 3**C**). OPLS-DA method is useful for preliminarily screening differential metabolites across cultivars or tissues[48]. The OPLS-DA model was evaluated using parameters R2X, R2Y, and Q2: R2X and R2Y represent the model's explanatory power for the X and Y matrices, respectively, while Q2 indicates predictive power. The closer R2X and R2Y are to 1, the more stable and reliable the model is. When $Q2 > 0.5$, it can be considered a valid model; if $Q2 > 0.9$, it is an excellent model[49]. In our study, the Q2 values of the four comparison groups were above 0.9, suggesting that the OPLS-DA models were stable. The OPLS-DA plots exhibited that the BSE by six different processes were separated, indicating significant differences among these four groups. The model's validity was confirmed through 200 permutation tests (Fig. 3**D**), yielding R2 and Q2 values of 0.3618 and −0.7203, respectively, suggesting that the model was not overfitted.

3.3. Identification of differential B. striata derived metabolites

Metabolites with $VIP > 1$ and $P\text{-value} < 0.05$ were identified as significantly different based on the OPLS-DA results [50]. To visualize the overall trends in metabolite content differences in BSE and the statistical significance of these differences across methods, the detected metabolites were displayed in a volcano plot (Fig. 3**E** and **Table S5**).

In the FBST versus DBST comparison, 967 differential metabolites were identified, including 105 up-regulated and 862 down-regulated metabolites (**Table S6**). A combined analysis revealed 703 differential metabolites for FBST vs. FBSC (**Table S7**), 735 for FBST vs. FBSH (**Table S8**), and 667 for FBSC vs. FBSH (**Table S9**).In the comparison between FBST and FBSC (**Table S7**), volcano plot classifies the differential metabolites into 16 categories, lipids (124, 17.63 %), flavonoids (75, 10.66 %), terpenoids (93, 13.22 %), and carbohydrates and their derivatives (55, 7.87 %) accounted for a relatively large proportion of the metabolites. Differential metabolite analysis revealed that among the 222 up-regulated metabolites, lipids (56,25.23 %) , flavonoids (5, 2.25 %)and terpenoids (35,15.77 %) accounted for the highest proportions. Among the 481 down-regulated metabolites, flavonoids (70,14.53 %), lipids (68, 14.14 %), and terpenoids (58, 11.79 %) were the most abundant categories. In FBST vs. FBSH (Table S8), lipid (106, 15.94 %), flavonoid (81, 12.18 %), terpenoid (79, 11.88 %), and carbohydrates and their derivatives (56, 8.4 %) accounted for a larger proportion. Among the 333 up-regulated metabolites, lipids (85, 25.68 %), terpenoids (56,8.4 %) and flavonoids (12,3.63 %) were predominant. Among the 402 down-regulated metabolites, flavonoids (78,19.40 %), terpenoids (23,5.72 %), lipids (21,5.2 %) and carbohydrates 41, 10.20 %) were the most abundant. Terpenoids and flavonoids are widely recognized as key bioactive components contributing to human health, as demonstrated by extensive research on their presence in ginkgo[51]. The findings indicated that ultrasonic-assisted crushing extraction, followed by both cold and hot water extraction methods, resulted in an increased yield of flavonoids and terpenoids.

A total of 667 differential metabolites were identified in FBSC vs. FBSH, with 416 up-regulated and 251 down-regulated (**Table S9**).

DBST vs. DBSC, DBST vs. DBSH, DBSC vs. DBSH differential metabolites are 670 (**Table S10**), 663 (**Table S11**), and 532 (**Table S12**), respectively. In DBST vs. DBSC, among the 599 up-regulated metabolites, lipids (143, 23.91 %), terpenoids (107, 17.86 %), and flavonoids (57, 9.53 %) were the most abundant. Among the 71 down-regulated metabolites, terpenoids (11, 15.49 %), lipids (11, 15.49 %) accounted for the highest percentage. In DBST vs. DBSH, 591 metabolites were up-regulated, primarily lipids (144, 24.4 %), terpenoids (105, 17.80 %), and flavonoids (58, 9.83 %). Among the 72 down-regulated metabolites, lipids (11, 15.49 %) and terpenoids (9, 12.68 %) accounted for higher percentages. A total of 532 differential metabolites were identified in DBSC vs. DBSH, with 170 up-regulated and 362 down-regulated (**Table S12**).

3.4. Screening of differential chemical components

The concentration of both free and bound phenolics exhibited considerable variation among radix *Puerariae thomsonii* varieties originating from different geographical locations[52]. Because of some non-experimental factors, *B. striata* from different origins can also affect the proportion of plant metabolites. We compared FBST from Yunnan with DBST from Bozhou, using $VIP > 2$, $P\text{-value} < 0.05$, and $Log_2FC > 2$ or $Log_2FC < 0.5$ as criteria to identify major differential metabolites. 27 metabolites are significantly overexpressed in DBST (**Table S13**, Fig. 4 and **Fig. S2-S3**), these compounds encompass a variety of types, including diterpenes, polyketides, flavonoids, carotenoid derivatives, phenylpropanoids, steroids, iridoids, and fatty acids. They are widely distributed across plants, fungi, and other natural organisms, exhibiting significant biological activities such as antioxidant, anti-inflammatory [53], antibacterial, anticancer[54], and endocrine regulatory functions.

B. striata polysaccharides (BSP) have been extensively studied. In our previous research, we developed an easy-to-swallow edible hydrogel by optimizing the mixture ratios of BSP at different concentrations with sodium hyaluronate [55]. This hydrogel presents a novel material for the food and pharmaceutical industries, illustrating how various extraction methods and concentrations can unlock new utilitarian values for the medicinal use of *B. striata*.

3.5. Metabolic pathway enrichment analysis

3.5.1. Metabolite differences between FBST vs. FBSC

Metabolic pathway enrichment analysis serves as a crucial method for elucidating the mechanisms underlying metabolic changes in biological samples[56]. In FBST vs. FBSC, we identified 703 differential metabolites. Conducting a KEGG enrichment analysis based on differential metabolites. We further scrutinized the key metabolic pathways and metabolite levels between these two groups. As the P-value associated with differential metabolic pathways decreased, the significance of pathway enrichment increased[57]. Each bubble represents a metabolic pathway. The difference in abundance (DA) scores could reflect the overall changes in all pathways [58]. The DA score plot indicated that all two pathways were downregulated in FBST vs. FBSC (Fig. 5**A**). The horizontal axes of the graph represent the DA score, while the vertical axes denote the KEGG metabolic pathway names. The DA score represents the cumulative changes of all metabolites within a specific metabolic pathway[59]. DA score of 1 signifies that the expression of all identified differential metabolites in that pathway is predominantly up-regulated, whereas DA score of −1 suggests a predominant down-regulation[60]. The length of the line segments corresponds to the absolute value of the DA score. KEGG enrichment analysis of *B. striata* extracted by different processes revealed biosynthesis pathways for flavonoids and isoflavonoids, which align with the metabolite results mentioned above. Flavonoids are among the most important and abundant compounds in *B. striata* and can be classified into various subclasses, including flavonoids, isoflavonoids, anthocyanins, proanthocyanidins, and gold ketones. These subclasses are categorized based on the degree of saturation, substitution, and oxidation of the C-ring [28]. For flavonoid biosynthesis, the levels of apigenin, acacetin, kaempferol, and kaempferol-3-O-glucoside were significantly lower in the FBST extract than in FBSC (Fig. 5**C**). Additionally, several metabolites involved in the biosynthesis of isoflavones—such as liquiritigenin, glyceollin II, calycosin, formononetin, rotenone, apigenin, ononin, and medicocarpin—were present at higher levels in the FBSC extract than in the FBST extract (Fig. 5**D**). Acacetin has demonstrated anti-inflammatory potential against NLRP3 inflammasomes [61]. Kaempferol ameliorates MC903-induced dermatitis by inhibiting type 2 inflammation[62]. Liquiritigenin, derived from licorice root, attenuates macrophage inflammation and collagen-induced arthritis in rats [63]. Therefore, compared to *B. striata* extracted using FBST, the FBSC extract exhibits higher antioxidant and antimicrobial activity.

3.5.2. Metabolite differences between FBST vs. FBSH

A total of 735 differential metabolites were identified between the FBST and FBSH groups. These metabolites were primarily enriched in the pathways of Stilbenoid, diarylheptanoid and Diterpenoid biosynthesis, Isoflavonoid biosynthesis, and Flavonoid biosynthesis (Fig. 5**B**), involving 5, 8, 13, and 15 metabolites, respectively (**Table S15**). The DA score map revealed that two pathways in FBST vs. FBSH were upregulated, while one pathway was downregulated (Fig. 5**B**). The isoflavone biosynthesis pathway included 13 metabolites, such as Naringenin, Daidzein, and Glyceollin II. The heatmap demonstrated that these 13 metabolites were significantly overexpressed in FBSH (Fig. 5**E**). Similarly, the flavonoid biosynthesis.

pathway included 15 metabolites, primarily liquiritigenin, butin, apigenin. A heatmap analysis reveals that these metabolites are highly expressed in FBSH (Fig. 5**F**). Naringenin prevents nonalcoholic steatohepatitis by modulating both the host metabolome and the gut microbiome in mice fed an MCD diet [64]. Soybean sapogenins ameliorate adriamycin-induced cardiac injury in rats by inhibiting both autophagy and apoptosis [65]. Butin attenuates memory impairment in streptozotocin-induced diabetic rats by inhibiting both oxidative stress and inflammatory responses [66]. Compared to FBST, FBSH plays a more significant role in promoting cardiovascular and nervous system resilience, as well as exerting physiological effects.

3.5.3. Metabolite differences between DBST vs. DBSC/DBST vs. DBSH

In the comparison between DBST and DBSC, we identified a total of 670 differential metabolites. Subsequently, we conducted a KEGG enrichment analysis. The results indicated that eight pathways were responsive to the metabolite differences between the DBST and DBSH groups. These pathways included Fatty acid degradation (4 metabolites) , Cutin, suberine and wax biosynthesis (6 metabolites), Biosynthesis of unsaturated fatty acids (7 metabolites), isoflavonoid biosynthesis (10 metabolites) , Flavonoid biosynthesis (10 metabolites), alpha-Linolenic acid metabolism (12 metabolites), Linoleic acid metabolism (13 metabolites), and Arachidonic acid metabolism pathway (23 metabolites) (**Fig. S4A** and **Table S16**), respectively. The DA score plot showed that both pathways were up-regulated in DBST vs. DBSC (**Fig. S4A**).

In the comparison between DBST and DBSH, 663 differential metabolites were identified. KEGG enrichment analysis revealed seven significantly affected metabolic pathways: Alpha-linolenic acid metabolism, Arachidonic acid metabolism, Isoflavonoid biosynthesis, Fatty acid degradation, biosynthesis of unsaturated fatty acids, Cutin, suberine, and wax biosynthesis, and Linoleic acid metabolism. These pathways involved 11, 21, 9, 4, 9, 7, and 13 metabolites, respectively (**Table S17**). The DA score plot indicated an upregulation of these pathways in DBSH compared to DBST (**Fig. S4B**). The biosynthesis pathways of flavonoids include 10 primary metabolites, such as liquiritigenin, naringenin, and afzelechin. A heatmap reveals that these metabolites are highly expressed in DBSC (**Fig. S4C**). 5-p-coumaroylquinic acid significantly influenced both the coloration and flavor profile of the tea infusion [67], in DBST vs. DBSC, we find 5-p-coumaroylquinic acid riched in DBSC, it may also influence the sensory attributes and flavor of DBSC. Similarly, the isoflavonoid biosynthesis pathways encompass 10 metabolites, including medicarpin, vestitone, and calycosin, which are also highly expressed in DBST, as indicated by the heatmap (**Fig. S4D**). Compared to DBST, DBSC may offer a better flavor. In contrast, another set of isoflavonoid biosynthesis pathways includes 9 metabolites, primarily vestitone, apigenin, and rotenone. The heatmap demonstrates that these metabolites are highly expressed in DBSH (**Fig. S4E**). Rotenone inhibits β-cell apoptosis and alleviates the effects of type 1 diabetes mellitus[68]. Vestitone, obtained from the heartwood of sandalwood, demonstrated antibacterial activity against cyanobacteria, with an inhibition zone of 16.62 ± 1.07 mm[69]. Apigenin, a biologically active flavonoid, demonstrates significant potential for the prevention and treatment of various diseases[70]. Our research indicates that apigenin, vestitone and rotenone is more abundantly expressed in the dried tubers of *B. striata*. This suggests that the direct comminution of these dried tubers may provide potential preventive health benefits.

Table 1
Bitter metabolites.

Compounds	Formula	Adducts	Retention time
Cinnamic acid	C9H8O2	M + H-H2O	4.0671
L-Phenylalanine	C9H11NO2	M + H-2H2O	4.0671
Naringenin	C15H12O5	M−H	6.723
Caffeic acid	C17H16O4	M−H	11.5959
P-Coumaric acid	C9H8O3	M + H-H2O, M + H	4.9257
Kaempferol	C15H10O6	M + H	4.4353
Kaempferol-3-O-Glucoside	C21H20O11	M−H	4.4202
Apigenin	C15H10O5	M + H, M + Na	5.8049
Cyanidin 3-(2G-glucosylrutinoside)	C33H41O20+	M + H	14.6699
4′-O-methyl-(−)-epicatechin-3′-O-beta-glucuronide	C23H26O12	M−H2O−H	4.6227
(−)-Epigallocatechin sulfate	C15H13O10S-	M + FA-H	1.4163
L-Histidine trimethylbetaine	C9H15N3O2	M + H	0.5673

3.6. Identification of the bitter compounds

B. striata, a medicinal plant rich in BSP, is notably characterized by its bitter taste. This study focuses on identifying the bitter metabolites present in BSE. Bitter substances are ubiquitous in plants and include polyphenols, alkaloids, amino acids and peptides, saponins, and inorganic salts[71]. The level of bitterness varies among plants depending on the type and concentration of bitter substances present[72]. Bitter metabolites are crucial in their pharmacological effects[73]. To identify bitter metabolites in BSE, we employed non-targeted metabolomics analysis. Subsequently, a combination of the BitterDB database and relevant literature[74,75] allowed us to identify 13 bitter metabolites (Table 1) as potential major contributors to BSE bitterness. Flavonoids and isoflavones in daylily have been shown to contribute to its flavor profiles, including bitterness, sourness, and astringency[49]. Flavonoids are responsible for the antidepressant, sleep-improving, and lactation-promoting effects of daylily[76,77]. The administration of cinnamic acid at doses of 45 and 90 mg/kg has been shown to reduce inflammation in acetic acid-induced colitis by inhibiting the activities of TNF-α, IL-6, and MPO, while also downregulating the expression of TLR-4 [78]. Kaempferol, at a concentration of 50 μM, significantly reduced the increased permeability of the monolayer barrier induced by high glucose in rat intestinal microvascular endothelial cells (RIMVECs). Moreover, it markedly attenuated the high glucose-induced thinning of the tight junction protein Claudin-5. Additionally, Kaempferol inhibited the VEGFR2/p38 pathway, thereby reducing angiogenesis and cell migration induced by high glucose [79]. Naringenin protects the kidney from ischemia/reperfusion (I/R) injury by reducing cell death and apoptosis, which is mediated via the activation of the Nrf2/HO-1 signaling pathway and the inhibition of ER stress [80]. Caffeic acid activates the Nrf2 signaling pathway to downregulate the TFR1 and ACSL4 genes while upregulating glutathione production, thereby helping to resist ferroptosis in the pMCAO rat brain and in oxygen-glucose deprivation/reoxygenation (OGD/R)-treated SK-N-SH cells in vitro [81]. The flavonoid compound apigenin, which is found naturally, exhibits a strong inhibitory effect on the proliferation of hepatocellular carcinoma, mediated by its ability to block microvesicle secretion[82].

We conducted a comparative analysis of the variations in bitterness-

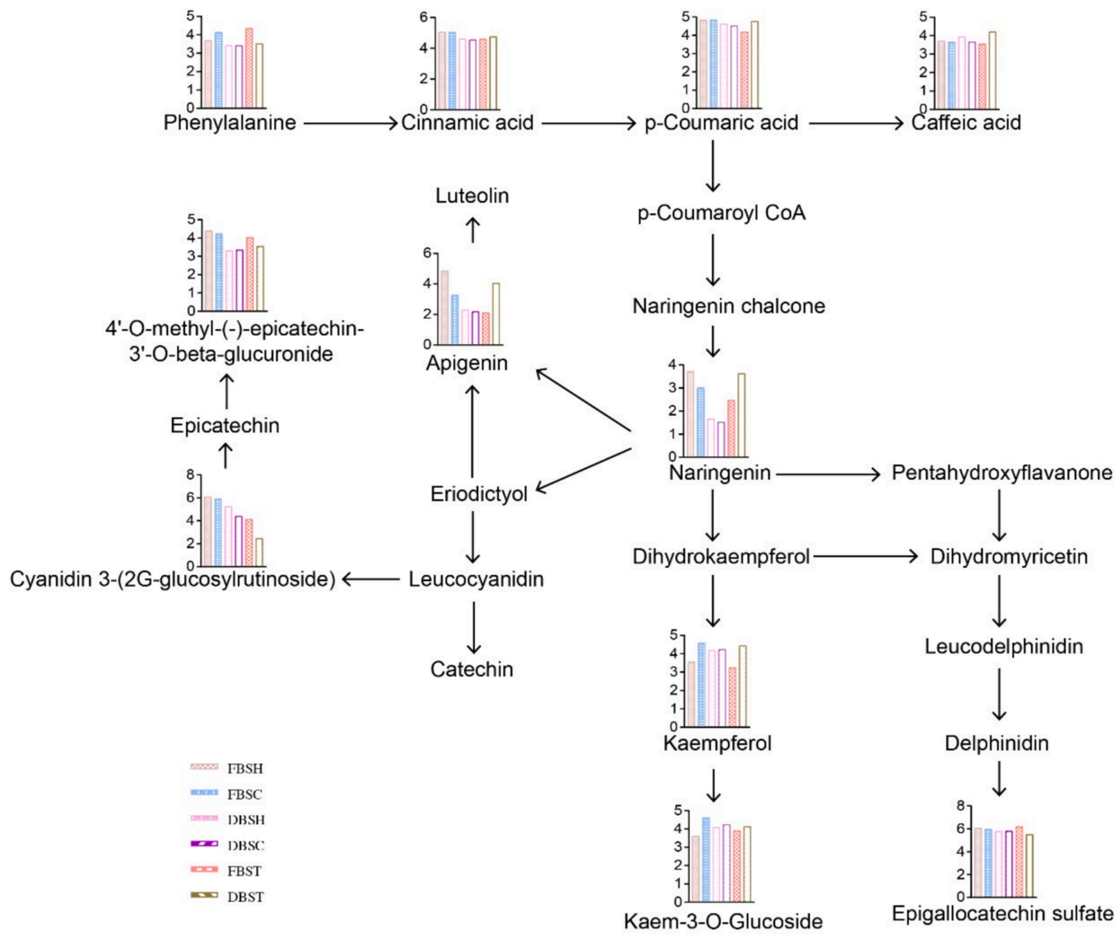

Fig. 6. A comparative analysis of the bitter metabolite biosynthetic pathways in *B. striata* extract was conducted under six distinct extraction conditions: FBSC, FBSH, FBST, DBSC, DBSH, and DBST. The y-axis represents the relative abundance of metabolites, quantitatively measured using UPLC-MS/MS. Bar graphs highlight the variations in specific flavonoids across each extraction condition.

related metabolites within the flavonoid biosynthetic pathways of FBSC, FBSH, FBST, DBSC, DBSH, and DBST. The changes in bitter metabolites during the cold and hot extraction processes of both dry and fresh *B. striata* were examined (Fig. 6). Histograms adjacent to the corresponding metabolites illustrate the variations in flavonoid and phenolic acid metabolite contents within the bitter metabolites of *B. striata* extracts across different procedures.

4. Conclusion

This study analyzed the metabolic profiles of *B. striata* extracts under six extraction methods: FBST, FBSH, FBSC, DBST, DBSH, and DBSC. A total of 1,945 metabolites were identified, categorized into primary and secondary classes, with lipids, flavonoids, and phenolic acids being the most abundant. Flavonoids, including apigenin, kaempferol, and naringenin, were highlighted for their antioxidant, anti-inflammatory, and wound-healing properties. Significant differences in metabolite profiles were observed between fresh and dried *B. striata* and across extraction methods. Thirteen bitter metabolites, such as cinnamic acid and kaempferol-3-O-glucoside, were identified as contributors to BSE's bitterness. KEGG pathway enrichment revealed notable changes in flavonoid and isoflavonoid biosynthesis, particularly in dried tubers under cold and hot extraction processes. In conclusion, *B. striata* extracts, rich in bioactive flavonoids and phenolic acids, demonstrate significant pharmacological potential. Optimized extraction methods enhance these metabolites, improving therapeutic efficacy and flavor.

CRediT authorship contribution statement

Juan Zhou: Writing – original draft, Validation, Formal analysis. **Yushen Feng:** Validation, Formal analysis. **Wenhao Zhou:** Formal analysis. **Mengying Zhang:** Formal analysis. **Fugui Liu:** Formal analysis. **Jian Mao:** Formal analysis. **Dajun Wu:** Formal analysis. **Yunpeng Cao:** Data curation. **Yigao Wu:** Data curation. **Lan Jiang:** Writing – review & editing, Writing – original draft, Validation, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Institutional Review Board Statement

Not applicable.

Informed Consent Statement

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ultsonch.2025.107266.

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